

REU-HPC: Week-5 Group Meeting

Group Meetings

Tuesday 3-4pm (ROC room 1.109)

<https://utexas.zoom.us/j/85978334209>

Week 5 Objectives (June 28 – July 5)

The primary goal for week 5:

- Run sample workloads with and without PEAK to collect profiling data and analyze the usage of mathematical libraries such as BLAS, LAPACK, and FFTW.
- Explore and create build scripts for more software packages from the Top 100 applications list by reviewing official documentation, searching for existing examples, and leveraging AI-assisted guidance.
- Meeting with HPC professionals: If you are interested, in the next few weeks, I can help arrange meetings with members of the HPC group so you can learn more about their work at TACC and hear about their experiences in the field of high-performance computing.

Useful information

- **New PEAK, readme, and documentation:**
Link: <https://github.com/peak-team/peak>
- Project directory: `/work2/05392/cylu/share/reu_2026/2.project`
 - **App build scripts:** abinit, berkeley_gw, dftb+, epw, gpaw, gromacs, lammmps, lapack, meep, milc, openblas, yambo
 - **Examples:** cp2k, dftb+, gromacs, lammmps, milc, ml, siesta, yambo
- **Frontera-top100-2025-1.txt:**
Link:
<https://drive.google.com/file/d/15VXVPnREFcz3hiVyVoCW8ThX5TO6WELO/view?usp=sharing>
- **package_list.xlsx**
Link:
<https://docs.google.com/spreadsheets/d/1JSHFZpOV8IA3M1mw86WJAuWZ8kotStPf/edit?usp=sharing&oid=101096215360805228479&rtpof=true&sd=true>